

Introduction to cloud computing.

Introduction :-

Cloud computing:-

- It is a technology that allows users to access computing resources over the internet.
- "cloud" refers to the remote server that store & manage data, located in data centers world wide.

1) Importance of cloud computing

① Growing trend:-

- cloud computing is continuously growing trend that provides better data storage, security, flexibility & collaboration.
- It also helps to improve decision-making.

② Market growth:-

- The cloud services market predicted to grow significantly.

③ Remote data storage:-

- Cloud storage involves storing data on remote servers that can be maintained, managed & accessed remotely.
- It offers secure data storage & accessibility from any device with permission.

④ Security & accessibility:-

- Even though data is easily accessible, the cloud ensures high level of safety & security.

⑤ media streaming benefits:-

- It provides cost-effective, flexible & secure solutions for media streaming companies.
- It supports both video on demand & live streaming services.



Study material provided by: Vishwajeet Londhe

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2] Characteristics

① On-demand self-service:-

→ Consumer can automatically get computing resources when needed.

② Ubiquitous network access:-

→ Resources are accessible over the internet on various devices & platforms.

③ Location-independent resource pooling:-

→ providers pool resources to serve multiple users & assign them based on demand.

④ Rapid elasticity:-

→ Resources can be quickly increased or decreased based on user needs.

⑤ Pay per use:-

→ Users are charged based on how much they use the service.

3] Pros & Cons of cloud computing.

Pros:-

① Lower computer costs:-

→ Application run in the cloud, reducing the need for high-end h/w

② Improved performance:-

→ fewer programs running on local machine lead to faster boot times & operation.

③ Instant s/w updates:-

→ Web-based apps always run the latest version automatically.

④ Unlimited storage capacity:-

→ cloud services offers nearly limitless storage option.

⑤ Increased data reliability:-

→ cloud storage is safer than local storage.

Cons

- ① constant internet connection required:-
→ cloud services depend on internet access.
- ② Feature limitations:-
→ Some cloud service may offer limited functionality.
- ③ Data security risk:-
→ Your data is stored online, making security concern.
- ④ Poor performance with slow connections:-
→ cloud services may not work well with low-speed internet.

Applications

- ① Business scalability:-
→ cloud helps online businesses meet demand spikes by quickly adding or provisioning resources.
- ② cost-efficient:-
→ Access powerful tools without extra h/w or s/w cost
- ③ Adaptable:-
→ Reach diverse devices & users.
- ④ Simplified:-
→ Access services without worrying about complexity.
- ⑤ Connected:-
→ share information smoothly with partners

A] Migrating in cloud computing.

Cloud migration:-

→ It is the process of moving data, application and business process from on-premise infrastructure to cloud environment.

Key aspects for successful migration:-

- Develop clear goal for business outcome
- Utilize skilled professionals to address core issues
- Use effective tools that are secure & user-friendly.

Focus areas:-

- ① Security:- control access, protect data in, transit & ensure backup.
- ② Governance:- meet compliance & security standards.
- ③ Cost management:- Control short & long term costs.
- ④ Scalability & resilience:- meet customer needs & grow with ease.

Benefits:-

Agility, flexibility, innovation, cost reduction, better IT management, scalability.

Migration process:-

- ① Plan:- Set clear business goal, assess systems and ensure visibility.
- ② Execute:- Migrate with minimal disruption and cost
- ③ Monitor:- Use tools to track & maintain the cloud environment.

Challenges:- Interoperability, resource availability, data integrity, network.

5] Seven step migration model.

- ① Explore cloud services.
- ② Cloud as a tool
- ③ cloud compatibility
- ④ Current cost.
- ⑤ Manage over operate.
- ⑥ Simplify process.
- ⑦ Gain knowledge.

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Step 1 :- conduct cloud migration assessment to identify issues at different levels.

Step 2 :- Isolate dependencies & components of your application that will need to be migrated.

Step 3 :- plan which part of the system should stay in your data center & which should go to the cloud.

Step 4 :- Redesign & reimplement parts of the application to fit cloud architecture.

Step 5 :- Use cloud features to improve your application

Step 6 :- Test the new cloud-based system thoroughly to make sure it works well.

Step 7 :- Based on result, make improvements and repeat the process until the migration is successful.

6] Trends in Computing

① IT maturity analysis :-

→ This compares different technologies based on how quickly they are adopted & the amount of current investment.

→ In cloud computing, trends include the adoption by small & ~~business~~ medium businesses & the use of hybrid cloud models.

② Technology trends to watch :-

① Virtualization :-

→ A technology that helps users create IT services using virtual resources instead of relying on physical h/w.

② Data growth :-

→ As data increases daily, organizations need more storage & processing power. ~~The~~

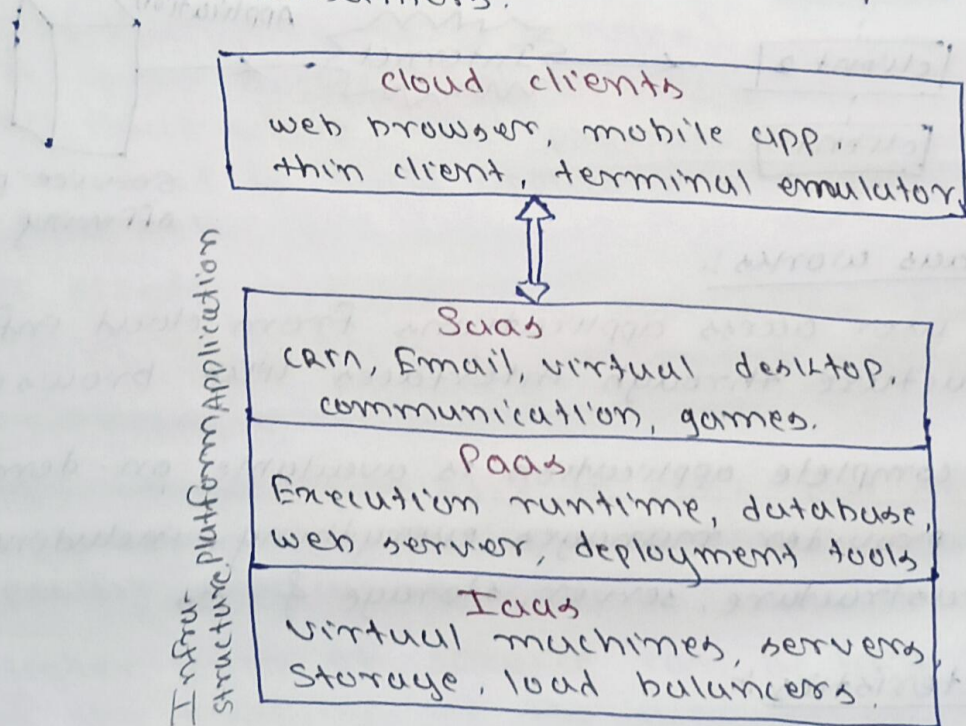
→ The cloud provides a solution to handle this data growth.

③ Green IT :-

→ This refers to using technology in a way that conserves resources & protects the environment.

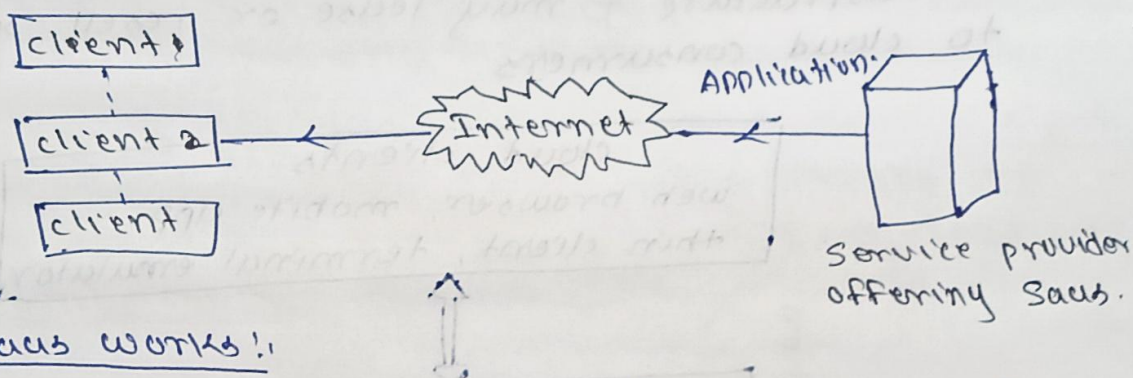
7] Cloud service model :-

- cloud service model explain what the cloud provider manages & what the client is responsible for.
- cloud services are provided over the internet rather than on-premises, making them scalable and easier to manage.
- cloud providers usually own & manage the infrastructure & may lease or resell services to cloud consumers.



SaaS (Software as a service)

- SaaS is cloud model where software is hosted and provided to users over the internet.
- In this model, the provider handles all patching and upgrades, while also keeping the infrastructure running.
- SaaS differs from the traditional sw distribution model, where user's buy & install sw on their own devices.



How SaaS works:

- The user access applications from cloud infrastructure through interfaces like browser.
- The complete application is available on-demand.
- The provider manages everything, including infrastructure, server, storage & OS.

Characteristics:

- SW applications or services are stored remotely.
- Users access these services via internet.
- No installation required on user's machine.
- Network based management allows centralized access to sw for multiple clients.
- SaaS provides one-to-many model.

Benefits

- ① Pay-per-use model :- client only pay for the services they use.
- ② Easier administration & invoicing.
- ③ Automatic updates & patch management.
- ④ Compatibility :- All users can access same version of slw
- ⑤ collaboration :- Easy to collaborate across diff. platforms.

PaaS (Platform as a service)

→ It is an application delivery model, also known as cloud-ware, that provides all the resources required to build applications & services via internet.

→ It offers resources for application design, development, testing, deployment, hosting & more.

Characteristics:-

- PaaS is closely related to SaaS but delivers a platform for development rather than a specific application.
- Higher levels of service can be built on top of the platform by the user.
- Customer can deploy application onto the cloud using tools supported by the provider.

Benefits:-

- Rapid allocation & deallocation of resources with a pay-as-you-go model.
- Reduced capital expenditure.
- On-demand resource availability.
- Supports team collaboration

IaaS (Infrastructure as a service)

- IaaS provides storage & infrastructure similar to in-house datacenters
- Key components include capacity, servers, storage & management tools.
- IaaS offers services such as server, space, network equipment, memory, CPU cycle & storage.
- Customer control operating systems, applications and storage as well as certain networking components.

Types of IaaS servers

- ① Physical server :- Dedicated h/w allocated for the customer's use.
- ② Dedicated virtual server :- virtual servers are allocated on a physical server.
- ③ Shared virtual server :- The customer access a virtual server on a shared physical device with other customer.

Advantages :-

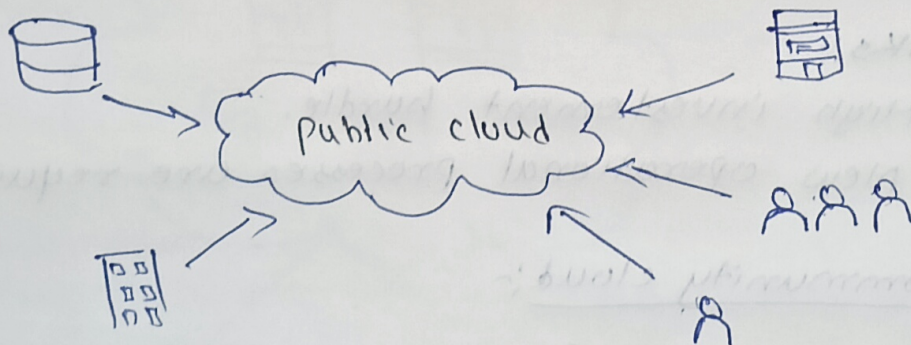
- H/w can scale easily
- Reduced h/w cost
- On-demand scaling
- Allows full management & administration of systems.
- Supports multi-tenant environments.

8] Cloud Deployment models :-

- cloud deployment model refers to the location & management of the cloud's infrastructure.
- They are defined by the ownership, control of architectural design, & degree of available customization.
- The main deployment models are private, public & community clouds.

① Public cloud :-

- cloud infrastructure is made available for general public & it is owned by an organisation that sells cloud services.
- public cloud involve large data centers that offer the same services to all users.
- They are accessible to everyone.



Benefits :-

- Low investment hurdle
- Good for testing / development.

Risks :-

- multi-tenancy & data transfer over internet increases security risk
- IT organization may have concern about losing control over data center management.

② Private cloud:-

- The cloud infrastructure is operated solely for a single organization
- It may be managed by organization itself or a third party.



Benefits:-

- fewer security concern
- The IT organization retains control over the data center.

Risks:-

- High investment hurdle.
- New operational processes are required.

③ Community cloud:-

- The cloud infrastructure is shared by several organization & supports a specific community that has shared concern.
- It may be managed by the organization or third party.

④ Hybrid cloud :-

→ The cloud infrastructure is a composition of two or more clouds that remain unique entities but are bound together by standardized technology that enable data and application portability.

Benefits

- Operational flexibility
- Scalability

Risks

- ~~Develop~~ Hybrid clouds are still being developed.
- Security concern.

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